## 3.28 Local Rings of Dimension 0

**Lemma 3.28.1 (Dimension 0 local ring)**
*Let $(A, \mathfrak{m})$ be a Noetherian local ring. Then the following are equivalent*

*a)* $\dim A = 0$ *(i.e. every prime ideal is maximal)*
*b)* $\mathfrak{m}^n = 0$ *for some* $n$
*c)* $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ *for some* $n$

*If $A$ is an integral domain then this is equivalent to $A$ being a field.*

*Proof.* a) $\implies$ b) By (3.21.6) $\mathfrak{r} := \sqrt{(0)} = \mathfrak{p}_1 \cap ... \cap \mathfrak{p}_n$ is a decomposition into minimal prime ideals. By assumption these are also maximal, and therefore by uniqueness $\sqrt{(0)} = \mathfrak{m}$ and the result follows from (3.3.48) as $\mathfrak{m}$ is finitely generated.

b) $\implies$ a) Let $\mathfrak{p}$ be a prime ideal then trivially $\mathfrak{m}^n \subset \mathfrak{p}$ so by (3.27.1) it is $\mathfrak{m}$-primary i.e. $\mathfrak{m}$ is a minimal prime of $\mathfrak{p}$ whence $\mathfrak{m} = \mathfrak{p}$.

b) $\iff$ c) One direction is obvious, and the converse follows from Nakayama's Lemma (3.20.5) because $\mathfrak{m}^n$ is finitely generated. ∎

© David Rufino CC BY-NC-ND 4.0